1	$1024+\ldots$	**B1**
	$\ldots+{}^{10}C_1\times\ldots x+{}^{10}C_2\times\ldots x^2+{}^{10}C_3\times\ldots x^3+\ldots$	**M1**
	$\ldots+10\times2^9\times\frac{3}{8}x+45\times2^8\times\left(\frac{3}{8}x\right)^2+120\times2^7\times\left(\frac{3}{8}x\right)^3$ Or two of $\ldots+1920x+1620x^2+810x^3$	**A1**
	$\ldots+1920x+1620x^2+810x^3$	**A1**
		(4)

2(a)		**B1**
		B1
		(2)
(b)	$h = 0.5$ seen or implied	**B1**
	$\text{Area} \approx \left\{\frac{0.5}{2}\right\} \times \left(1.319 + 2(\text{"1.075"} + 1.001 + \text{"1.041"} + 1.223) + 1.850\right)$	**M1**
	$=$ awrt 2.96	**A1**
		(3)
(c)	$3 + \log_{10}\left(\sin(x)\right) = 4 - \left(1 - \log_{10}\left(\sin(x)\right)\right)$	**B1**
	$\int_{0.5}^{3} 3 + \log_{10}(\sin(x))\,\mathrm{d}x = \left[4x\right]_{0.5}^{3} - \text{"2.96"} = 12 - 2 - \text{"2.96"}$	**M1**
	$= 7.04$ follow through $10 -$ their 2.96	**A1ft**
		(3)
Alt (c)	$\int_{0.5}^{3} \log_{10}(\sin(x))\,\mathrm{d}x = \int_{0.5}^{3} 1\,\mathrm{d}x - \text{"2.96"} = 2.5 - \text{"2.96"}$	**B1**
	$\int_{0.5}^{3} 3 + \log_{10}(\sin(x))\,\mathrm{d}x = \left[3x\right]_{0.5}^{3} + \left(2.5 - \text{"2.96"}\right) = 9 - 1.5 + 2.5 - \text{"2.96"}$	**M1**
	$= 7.04$ follow through $10 -$ their 2.96	**A1ft**
		(3)

x	0.5	1	1.5	2	2.5	3
y	1.319	**1.075**	1.001	**1.041**	1.223	1.850

3(i)	E.g. $n = 1: \ 2^3 - 1^3 = 7$, $n = 2: 3^3 - 2^3 = 19$, $n = 3: \ 4^3 - 3^3 = \ldots$ Or identifies counterexample directly.	**M1**
	e.g. $6^3 - 5^3 = 91 = 7 \times 13$ so not true for $n = 5$, hence statement is not true.	**A1**
		(2)

Way 1	$m_{AC} = \frac{-6-0}{7-1} = ...$ or $m_{BC} = \frac{-6-(-10)}{7-3} = ...$	**M1**
	$m_{AC} = -1$ and $m_{BC} = 1$	**A1**
	So $m_{AC} \times m_{BC} = -1 \times 1 = -1$ or e.g. m_{AC} is negative reciprocal of m_{BC}	**dM1A1**
	So e.g. angle (at C) is a right angle hence AB is a diameter (Or equivalent)	**A1**
		(5)
Way 2	$AB = \sqrt{(3-1)^2 + (-10-0)^2} = ...$ or $AC = \sqrt{(7-1)^2 + (-6-0)^2} = ...$ or $BC = \sqrt{(7-3)^2 + (-6+10)^2} = ...$	**M1**
	$AB = \sqrt{104}\left(2\sqrt{26}\right), AC = \sqrt{72}\left(6\sqrt{2}\right), BC = \sqrt{32}\left(4\sqrt{2}\right)$	**A1**
	$AB^2 = 104 = 72 + 32 = AC^2 + BC^2$	**dM1A1**
	Hence ABC is a right-angle triangle with hypotenuse AB hence AB is a diameter. (Or equivalent)	**A1**
		(5)

Way 3	If AB is diameter centre must be midpoint of AB ie $M\left(\frac{1+3}{2},\frac{0-10}{2}\right)$	**M1**
	$=(2,-5)$	**A1**
	$MA=\sqrt{(2-1)^2+(-5-0)^2}=\sqrt{26},\ MC=\sqrt{(2-7)^2+(-5-(-6))^2}=\sqrt{26}$	**dM1A1**
	$MA=\sqrt{26}, MC=\sqrt{26}$ so $MA=MC(=MB)$ As the length from M to each of A and C is the same M is the centre of the circle hence AB is a diameter. (Or equivalent)	**A1**
		(5)

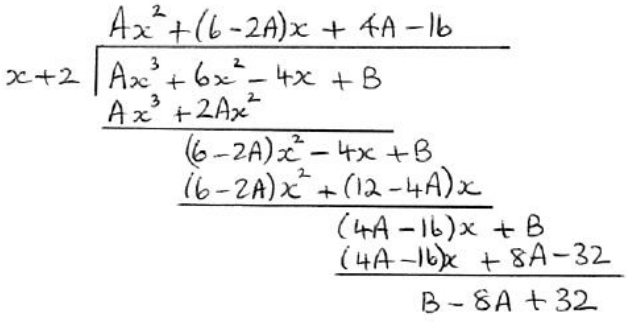
Way 4	If AB is diameter centre must be midpoint of AB ie $M\left(\frac{1+3}{2},\frac{0-10}{2}\right)$	**M1**
	$=(2,-5)$	**A1**
	$MA=r=\sqrt{(2-1)^2+(-5-0)^2}=\sqrt{26}$ $\Rightarrow (x-2)^2+(y+5)^2=26$ $C(7,-6)\Rightarrow (7-2)^2+(-6+5)^2=5^2+1^2=26$	**dM1A1**
	As C also satisfies the equation of the circle then AB must be the diameter (or equivalent) There must be some further justification as above rather than just "AB is a diameter" which may be in a preamble e.g. If C lies on the circle…	**A1**
		(5)

4	$\log_4 a+\log_4 b=\log_4 ab$ or $\log_4 "ab"=3\Rightarrow "ab"=4^3$	**M1**
	$ab=K, a-b=8\Rightarrow a(a-8)=K$ or $(b+8)b=K$	**dM1**
	$a^2-8a-64=0$ or $b^2+8b-64=0$	**A1**
	$a=\frac{-(-8)+\sqrt{(-8)^2-4\times1\times-64}}{2\times1}=...$ or $b=\frac{-8+\sqrt{8^2-4\times1\times-64}}{2\times1}=...$	**M1**
	$a='4+4\sqrt{5}'\Rightarrow b=-4+4\sqrt{5}$	**M1**
	$a=4+4\sqrt{5}$ and $b=-4+4\sqrt{5}$ and no other solutions.	**A1**
		(6)

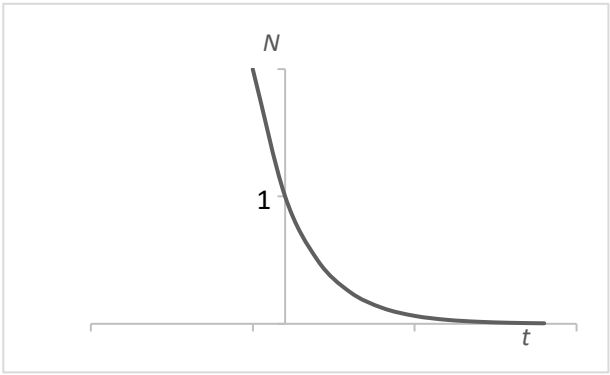
5	$3\tan(\theta+43°)=2\cos(\theta+43°)\Rightarrow 3\dfrac{\sin(\theta+43°)}{\cos(\theta+43°)}=2\cos(\theta+43°)$ $\Rightarrow 3\sin(\theta+43°)=2\cos^2(\theta+43°)$	**M1**
	$\Rightarrow 3\sin(\theta+43°)=2\left(1-\sin^2(\theta+43°)\right)$	**M1**
	$\Rightarrow 2\sin^2(\theta+43°)+3\sin(\theta+43°)-2=0$ $\Rightarrow \left(2\sin(\theta+43°)-1\right)\left(\sin(\theta+43°)+2\right)=0\Rightarrow \sin(\theta+43°)=...$	**M1**
	$\sin(\theta+43°)=\dfrac{1}{2}$	**A1**
	$\theta=\arcsin\text{"}\dfrac{1}{2}\text{"}-43°$	**M1**
	$\theta=-13°,\ 107°$	**A1**
		(6)

6(a)	$u_n = ar^{n-1} \Rightarrow ar + ar^2 = 6$ and $ar^3 = 8$	**M1**
	$\Rightarrow \frac{ar + ar^2}{ar^3} = \frac{6}{8} \Rightarrow 1 + r = \frac{3}{4}r^2$	**M1**
	$\Rightarrow 3r^2 - 4r - 4 = 0$*	**A1***
		(3)
(a) Way 2	$u_4 = 8 \Rightarrow u_3 = \frac{8}{r},\ u_2 = \frac{8}{r^2}$	**M1**
	$\frac{8}{r} + \frac{8}{r^2} = 6$	**M1**
	$\Rightarrow 3r^2 - 4r - 4 = 0$*	**A1***
		(3)
(b)	$r = -\frac{2}{3}$	**B1**
	$ar^3 = 8 \Rightarrow a = \frac{8}{\left(-\frac{2}{3}\right)^3} = ...$	**M1**
	$u_1 = -27$	**A1ft**
		(3)
(c)	$S_\infty = \frac{-27}{1 - \left(-\frac{2}{3}\right)} = ...$	**M1**
	$= -\frac{81}{5}$	**A1**
		(2)

7	$(x+2)$ a factor $\Rightarrow \mathrm{f}(-2)=0 \Rightarrow -8A+24+8+B=0$	**M1A1**
	$\int \mathrm{f}(x)\mathrm{d}x = \frac{A}{4}x^4+2x^3-2x^2+Bx$	**M1** **A1**
	$\int_3^5 \mathrm{f}(x)\mathrm{d}x = 176 \Rightarrow \left[\frac{A}{4}x^4+2x^3-2x^2+Bx\right]_3^5 = 176$ $\Rightarrow \left(\frac{A}{4}5^4+2(5^3)-2(5^2)+5B\right)-\left(\frac{A}{4}3^4+2(3^3)-2(3^2)+3B\right)=176$	**dM1**
	$\left.\begin{array}{r} 8A-B=32 \\ 136A+2B=12 \end{array}\right\} \Rightarrow A=\ldots, B=\ldots$	**dM1**
	$A=\frac{1}{2},\ B=-28$	**A1**
		(7)

$f(-2)=0 \Rightarrow -8A+24+8+B=0 \Rightarrow B=8A-32$	**M1A1**
$\int f(x)dx = \frac{A}{4}x^4+2x^3-2x^2+8Ax-32x$	**M1A1**
$\Rightarrow \left(\frac{A}{4}5^4+2(5^3)-2(5^2)+40A-160\right)-\left(\frac{A}{4}3^4+2(3^3)-2(3^2)+24A-96\right)=176$	**dM1**
$\Rightarrow A=...,\ \ B=...$	**dM1**
$A=\frac{1}{2},\ B=-28$	**A1**
Long Division for reference:  Score the first M1 for a complete method to obtain a remainder in terms of A and B that is set = 0 and A1 for a correct equation e.g. $B - 8A + 32 = 0$	

8(a)	$\left(\frac{\mathrm{d}y}{\mathrm{d}x}=\right)Ax^3+B+\frac{C}{x^3}$	**M1**
	$\left(\frac{\mathrm{d}y}{\mathrm{d}x}=\right)4\times256x^3-304-2\times27x^{-3}$ oe e.g. $1024x^3-304-\frac{54}{x^3}$	**A1A1**
		(3)
(b)	$\frac{\mathrm{d}y}{\mathrm{d}x}=0$ at SP $\Rightarrow 1024x^3-304-\frac{54}{x^3}=0\Rightarrow 1024x^6-304x^3-54=0$	**M1**
	$\Rightarrow 512\left(x^3\right)^2-152x^3-27=0\Rightarrow\left(8x^3+1\right)\left(64x^3-27\right)=0\Rightarrow x^3=...$	**dM1**
	$x^3=-\frac{1}{8}$ or $x^3=\frac{27}{64}$ or $x=-\frac{1}{2}$ or $x=\frac{3}{4}$	**A1**
	$y=256\left("-\frac{1}{2}"\right)^4-304\left("-\frac{1}{2}"\right)-35+\frac{27}{\left("-1/2"\right)^2}=...$ or with $x=\frac{3}{4}$	**dM1**
	Coordinates are $\left(-\frac{1}{2},241\right)$ and $\left(\frac{3}{4},-134\right)$	**A1**
		(5)

9(a)		
	Correct shape from or passing through a point on positive vertical axis. May extend to the left of the vertical axis and allow to pass into quadrant 4. There must be no (obvious) turning points. Labels not required on axes and ignore any that are given.	**M1**
	Shape and position correct, accept 1 or k as intercept on the positive vertical axis and allow to extend to the left of the vertical axis as shown. Condone (1, 0) or (k, 0) as long as it is in the correct position. The curve should approach a horizontal asymptote that is half-way between the horizontal axis and the intercept or below. Be tolerant with "wobbles" as it approaches the asymptote. May just "touch" the horizontal axis but not go below it. Labels not required on axes and ignore any that are given.	**A1**
		(2)
(b)	$\frac{1}{2}k = k\lambda^{5700} \Rightarrow \lambda^{5700} = \frac{1}{2}$	**M1**
	$\Rightarrow \lambda = \left(\frac{1}{2}\right)^{\frac{1}{5700}} = 0.999878$ to 6 d.p.*	**A1***
		(2)
(c)	When $t = 3250$, $N = 15 \times 0.999878^{3250} = ...$	**M1**
	$=$ awrt 10.1 (grams)	**A1**
		(2)
(d)	$18 = 25 \times 0.999878^{t}$	**B1**
	$\Rightarrow 0.999878^{t} = \frac{18}{25} \Rightarrow t = \frac{\log \frac{18}{25}}{\log 0.999878} = ...$	**M1**
	$t = 2692.49...$ so item is 2700 years old	**A1**
		(3)

10(a)	Equation of circle is $(x-3)^2+(y-5)^2=r^2$ and line is $y=2x+k$ So intersect when $(x-3)^2+(2x+k-5)^2=r^2$	**M1**
	$\Rightarrow x^2-6x+9+4x^2+4(k-5)x+(k-5)^2=r^2$ $\Rightarrow 5x^2+(-6+4k-20)x+9+k^2-10k+25-r^2=0$	**dM1**
	$\Rightarrow 5x^2+(4k-26)x+k^2-10k+34-r^2=0$*	**A1***
		(3)
(b)	Tangent to $C \Rightarrow b^2-4ac=0 \Rightarrow (4k-26)^2-4\times5\times(k^2-10k+34-r^2)=0$	**M1**
	$\Rightarrow 16k^2-208k+676-20k^2+200k-680+20r^2=0$ $\Rightarrow 5r^2=...$	**M1**
	$\Rightarrow 5r^2=k^2+2k+1=(k+1)^2$	**A1**
		(3)
(b) **Way 2**	Gradient of BX is $-\frac{1}{2}$ so equation of BX is $y-5=-\frac{1}{2}(x-3)$ $y-5=-\frac{1}{2}(x-3), y=2x+k \Rightarrow x=..., y=...\left(\frac{13-2k}{5}, \frac{26+k}{5}\right)$	**M1**
	$\left(\frac{13-2k}{5}-3\right)^2+\left(\frac{26+k}{5}-5\right)^2=r^2$	**dM1**
	$\Rightarrow 5r^2=k^2+2k+1=(k+1)^2$	**A1**
		(3)
(c)	Triangle AXB is right angled so $AB^2+r^2=XA^2=(3-0)^2+(5-k)^2$	**M1**
	$AB^2=4r^2$ so $AB^2+r^2=5r^2$	**M1**
	$\Rightarrow 5r^2=9+(5-k)^2$	**A1**
	$\Rightarrow k^2+2k+1=9+25-10k+k^2$	**M1**
	$\Rightarrow 12k=33 \Rightarrow k=...$	**dM1**
	$k=\frac{11}{4}$	**A1**
		(6)

Alt (c)	At B $x=-\dfrac{b}{2a}=\dfrac{13-2k}{5}$, $y=2\left(\dfrac{13-2k}{5}\right)+k$	**M1**
	$AB^2=\left(\dfrac{13-2k}{5}-0\right)^2+\left(\dfrac{26-4k}{5}+k-k\right)^2$	**M1 A1**
	$\Rightarrow 25\times 4r^2=\left(13-2k\right)^2+(26-4k)^2=\ldots$	**M1**
	$\Rightarrow 20\times(k+1)^2=\left(13-2k\right)^2+(26-4k)^2=20k^2-260k+845$ $\Rightarrow k^2+2k+1=k^2-13k+\dfrac{169}{4}\Rightarrow k=\ldots$	**dM1**
	$\Rightarrow k=\dfrac{11}{4}$	**A1**
		(6)

Way 5	If AB is diameter centre must be midpoint of AB ie $M\left(\frac{1+3}{2},\frac{0-10}{2}\right)$	**M1**
	$=(2,-5)$	**A1**
	Attempts 2 of: $m_{BC}=\frac{-6+10}{7-3}=1$ and midpoint is $\left(\frac{7+3}{2},\frac{-6-10}{2}\right)=(5,-8)$ so perpendicular bisector is $y+"8"=-\frac{1}{"1"}(x-"5")$ or $m_{AC}=\frac{7-1}{-6-0}=-1$ and midpoint is $\left(\frac{7+1}{2},\frac{-6}{2}\right)=(4,-3)$ so perpendicular bisector is $y+"3"=-\frac{1}{"-1"}(x-"4")$ or $m_{AB}=\frac{-10-0}{3-1}=-5$ and midpoint is $\left(\frac{3+1}{2},\frac{-10}{2}\right)=(2,-5)$ so perpendicular bisector is $y+"5"=-\frac{1}{"-5"}(x-"2")$ $y+8=-(x-5)$ oe or $y+3=x-4$ oe or $y+5=\frac{1}{5}(x-2)$ oe **And solves simultaneously:** E.g. $y+3=x-4, y+8=5-x\Rightarrow 5-x-5=x-4\Rightarrow x=2, y=-5$	**dM1**
	Hence centre of circle is $(2,-5)$	**A1**
	E.g. Midpoint of AB is the centre of the circle so AB is a diameter (or equivalent)	**A1**
		(5)

Way 6	Uses $(x-a)^2+(y-b)^2=r^2$ With (1, 0), (7, –6) and (3, –10) To find $(a, b) = \ldots$ or $r/r^2 = \ldots$	**M1**
	Centre (2, –5) or radius $\sqrt{26}$	**A1**
	E.g. Equation of AB is $y=-5(x-1)$ and $-5(2-1)=-5$ or $AB=\sqrt{(3-1)^2+(-10-0)^2}=\sqrt{104}=2\sqrt{26}$ or midpoint of AB is $\left(\frac{1+3}{2},\frac{0-10}{2}\right)=(2,-5)$	**dM1A1**
	So centre is on AB or AB is twice the radius or midpoint is the centre hence AB is a diameter of the circle. (or equivalent)	**A1**
		(5)

(ii) **Way 7**	If AB is diameter centre must be midpoint of AB ie $M\left(\frac{1+3}{2},\frac{0-10}{2}\right)$	**M1**
	$=(2,-5)$	**A1**
	Circle centre C radius r is $(x-7)^2+(y+6)^2=r^2$ Circle centre B radius r is $(x-3)^2+(y+10)^2=r^2$ These intersect when $(x-7)^2+(y+6)^2=(x-3)^2+(y+10)^2$ $\Rightarrow x+y=-3$ Circle centre A radius r is $(x-1)^2+y^2=r^2$ $(x-3)^2+(y+10)^2=(x-1)^2+y^2 \Rightarrow x-5y=27$	**dM1A1**

	Solves simultaneously: $x+y=-3,\ x-5y=27 \Rightarrow x=2, y=-5$	
	E.g. Midpoint of *AB* is the centre of the circle so *AB* is a diameter (or equivalent)	**A1**
		(5)